

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 19: Propellers & Constant Speed Governors

Syllabus Source: CAE Oxford (pp. 93-138 (Pwr)) | Pass Target: $\geq 90\%$ AviationExam

1. PROPELLER AERODYNAMICS: BLADE ANGLE VS ANGLE OF ATTACK

A propeller blade is a rotating airfoil that converts engine shaft torque into forward aerodynamic thrust:

- **Blade Element Theory:** The blade is twisted from root to tip. Because rotational speed ($\omega \times r$) increases towards the tip while forward airspeed is constant, the blade angle must decrease progressively toward the tip to maintain an optimal aerodynamic Angle of Attack (α) along the entire blade length.
- **Blade Angle (Pitch):** The acute angle between the blade chord line and the propeller plane of rotation.
- **Angle of Attack (α):** The angle between the blade chord line and the RELATIVE AIRFLOW (which is the vector sum of rotational velocity and forward airspeed).
- **Propeller Slip:** Geometric Pitch (theoretical distance advanced in one revolution with zero slip) minus Effective Pitch (actual distance advanced in flight). $\text{Slip} = \text{Geometric Pitch} - \text{Effective Pitch}$.

2. CONSTANT SPEED UNIT (CSU / GOVERNOR) OPERATION

A Constant Speed Propeller maintains an exact, selected engine RPM automatically regardless of aircraft airspeed or throttle position, modulating blade pitch hydraulically via an engine-driven governor (CSU):

Flight Condition	Governor Flyweights & Valve State	Propeller Blade Pitch Response
ON-SPEED	Centrifugal force on rotating flyweights EXACTLY BALANCES speeder spring tension.	Pilot valve lands cover oil metering ports; oil trapped in propeller cylinder; blade pitch remains constant.
OVER-SPEED (Dive / Throttle Advance)	Rotational RPM increases $\rightarrow$ flyweights tilt OUTWARDS against spring, lifting pilot valve.	Meters high-pressure engine oil to/from propeller hub to COARSE (increase) blade pitch. Increased aerodynamic load slows engine back down to selected RPM!
UNDER-SPEED (Climb / Throttle Retard)	Rotational RPM drops $\rightarrow$ speeder spring overcomes flyweights and pushes pilot valve DOWN.	Changes oil flow to FINE (decrease) blade pitch. Reduced aerodynamic drag allows engine to accelerate back to selected RPM!

3. FEATHERING SYSTEMS & WINDMILLING DRAG

If a multi-engine aircraft suffers an engine failure in flight, a windmilling propeller creates massive parasite drag and severe yaw:

- **Feathering (Blade Angle $\sim 85^\circ$ to 90°):** Blades are aligned parallel to the relative airflow, stopping propeller rotation and minimizing aerodynamic drag to the lowest possible value.
- **Feathering Mechanism:** Multi-engine propellers use heavy counterweights and internal feathering springs to drive blades into feather. Oil pressure opposes the springs to move blades toward fine pitch. Therefore, if engine oil pressure is lost, the propeller FEATHERS AUTOMATICALLY for safety!

- Centrifugal Latch Pins: Prevent the propeller from feathering when the engine is shut down on the ground below ~800-1,000 RPM. (If blades feathered on the ground, starting loads on the electric starter motor would be dangerously excessive).
- Unfeathering Accumulator: Stores high-pressure oil during normal flight. When the pilot moves the propeller lever out of feather in flight, an unfeathering solenoid releases accumulator oil back to the prop cylinder, driving blades back to fine pitch to windmill and air-start the engine.

[!] EXAM TRAP (AVIATIONEXAM) - Single-Engine vs Multi-Engine Propeller Pitch Failure Modes

- Single-Engine Propeller: Engineered to fail toward FINE PITCH (low pitch) upon oil loss, ensuring maximum power remains available for safe landing! - Multi-Engine Propeller: Engineered to fail toward FEATHER (high pitch) upon oil loss (driven by counterweights and springs), preventing catastrophic windmilling drag and uncontrollable asymmetric yaw!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Constant Speed Governor & Levers

1. Governor Action: - Overspeed: Flyweights fly outward -> Pilot valve directs oil to increase blade angle -> Engine slows down. - Underspeed: Speeder spring overcomes flyweights -> Blade angle decreases -> Engine speeds up. 2. Golden Rule of Engine Levers: - Increasing Power: PROPELLER lever first (high RPM), then THROTTLE lever (high MAP). - Decreasing Power: THROTTLE lever first (lower MAP), then PROPELLER lever (lower RPM). 3. Feathering: Blade angle turns to 85-90 degrees parallel to airflow, stopping windmilling.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In a multi-engine aircraft equipped with constant speed propellers, what happens to blade pitch if engine oil pressure drops?

- [A]: The propeller drives to full reverse pitch
- [B]: The propeller drives automatically to the FEATHERED position under the action of mechanical counterweights and feathering springs
- [C]: The propeller remains locked in fine pitch
- [D]: The propeller falls off

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Multi-engine propellers use oil pressure to maintain fine pitch. If oil pressure drops to zero, internal springs and counterweights drive the blades to full feather (~90°) to prevent high windmilling drag.

Q2: In a constant speed propeller governor (CSU), what directly balances the centrifugal force generated by the flyweights?

- [A]: Propeller blade torque
- [B]: The tension of the pilot-controlled speeder spring
- [C]: Atmospheric ambient air pressure
- [D]: High-pressure nitrogen

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The speeder spring tension (controlled by the cockpit blue propeller lever) pushes down on the flyweights. Flyweights spin with engine RPM and tilt outward with centrifugal force, balancing at 'on-speed'.

Q3: Why are propeller blades manufactured with an aerodynamic twist from root to tip?

- [A]: To make the blades stronger near the tip
- [B]: To maintain a relatively constant aerodynamic Angle of Attack along the entire length of the blade as rotational velocity increases toward the tip
- [C]: To reduce manufacturing cost
- [D]: To improve ground clearance

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Linear rotational speed is lowest at the hub and fastest at the blade tip. To prevent tip stall and maintain uniform lift distribution, blade angle is twisted (highest at root, lowest at tip).